

Plant Biotechnology

Engineering crops for better yield, resilience, and nutrition

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Plant Biotechnology applies genetic and biological techniques to improve crop traits, developing plants with better yield, disease resistance, or nutritional value.

- DEFINITION

The application of biotechnology techniques to modify or improve plants for agricultural, industrial, or pharmaceutical purposes.

- WHY IT MATTERS

Plant biotechnology has enabled crops that resist pests and disease, tolerate drought, and produce higher yields, playing a major role in global food security.

- WORKING PRINCIPLE

INPUT / PROBLEM

CORE PROCESS

OUTCOME / APPLICATION

Plant biotechnologists introduce or modify specific genes in plants to confer desired traits, using techniques ranging from traditional breeding assistance to precise gene editing tools like CRISPR.

- QUICK FACTS

- Golden Rice was engineered to address vitamin A deficiency, a major cause of childhood blindness in developing countries
- CRISPR-edited crops can be developed significantly faster than through traditional breeding

- KEY TERMS

Genetic modification

CRISPR

Tissue culture

Biofortification

- CORE CONCEPTS

- Genetic modification techniques
- Tissue culture propagation
- Marker-assisted breeding
- CRISPR gene editing in plants

- APPLICATIONS

- Pest and disease-resistant crop development
- Drought-tolerant crop varieties
- Nutritionally enhanced crops
- Rapid plant propagation via tissue culture

- ADVANTAGES

- Improves crop yield and resilience
- Can enhance nutritional content of staple foods
- Reduces reliance on chemical pesticides in some cases

- LIMITATIONS

- Public acceptance of genetically modified crops varies widely
- Regulatory approval processes can be lengthy
- Risk of reduced genetic diversity in widely adopted varieties

- CAREER OPPORTUNITIES

Plant biotechnologist, Crop scientist, Genetic engineer, Agricultural research scientist

- RESEARCH AREAS

CRISPR crop editing, Climate-resilient crop traits, Biofortification

- INDUSTRY APPLICATIONS

Seed companies, Agricultural biotechnology, Food security programs, Agrochemical industry

- FUTURE SCOPE

CRISPR gene editing is enabling faster, more precise crop improvement compared to traditional genetic modification methods.

- CURRENT TRENDS

Growing use of CRISPR gene editing to develop crop traits faster and with greater precision than earlier genetic modification methods.

- SUMMARY

Plant biotechnology engineers crops with improved yield, resilience, and nutrition, playing a major role in global food security.